NATIONAL BATTERY SUPPLY

844-539-2555 (Toll Free)

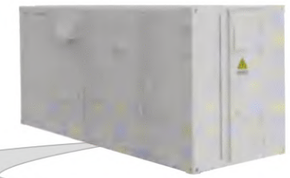
Sales@NationalBatterySupply.com

5416 W Sligh Ave. Suite 106 Tampa FL. 33634

OEM private label high energy density and deep cycle battery manufacture.

NATIONAL BATTERY SUPPLY

INDUSTRY

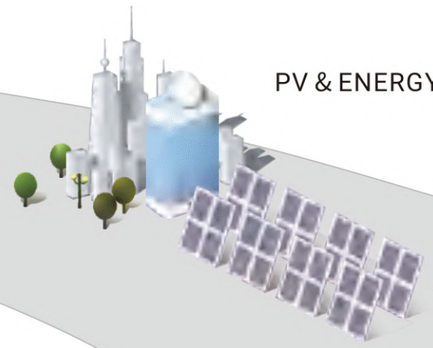




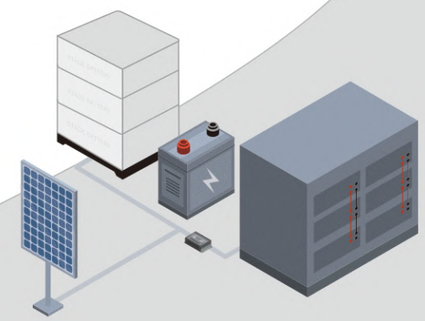
PV OFF-GRID SYSTEM

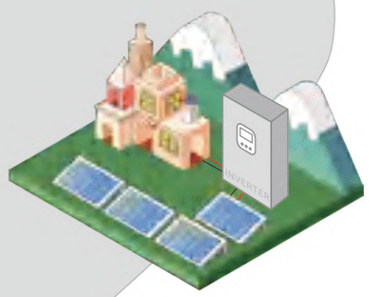


PV & ENERGY STORAGE SYSTEM



LITHIUM BATTERY PACK





SOLAR INVERTER

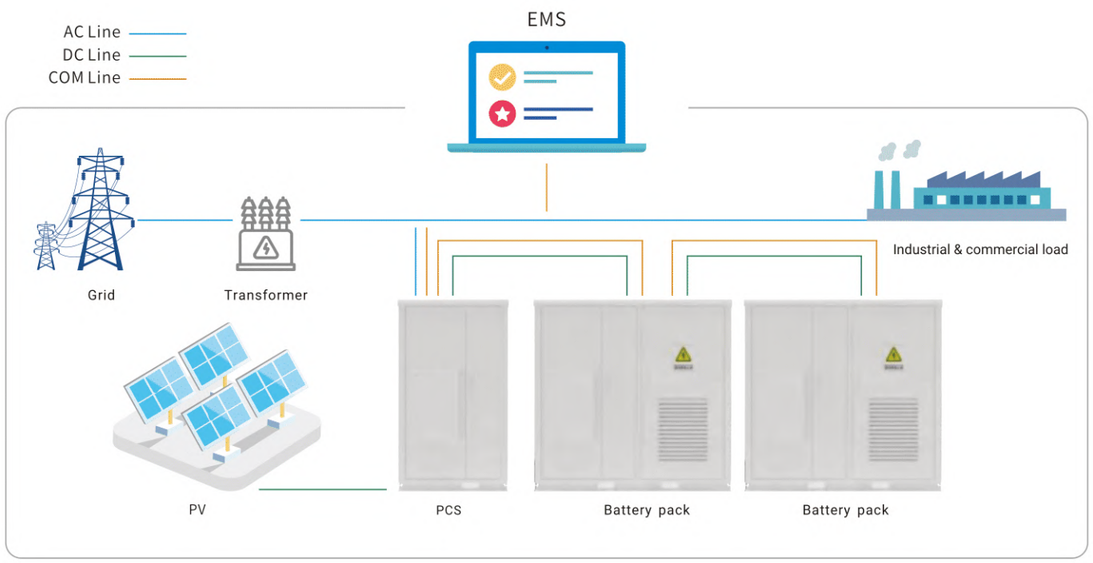
PORTABLE MOBILE POWER SUPPLY





INDUSTRIAL & COMMERCIAL ENERGY STORAGE

Power grid | UPS | Intelligent control | Energy saving | Green power



NBS
NATIONAL BATTERY SUPPLY





Technical Parameters

Cell parameters			
Cell type	Lithium iron phosphate(LFP) 280Ah		
Rated voltage	3.2V		
Depth of discharge	Deep discharge 90%		
Installed capacity			
Rated power	215KWh	604.8KWh	1MWh
Battery rated voltage	768V	720V	720V
Battery voltage range	600V-876V	560V-820V	560V-820V
DC discharge current	140A	140A	140A
Single Cluster Battery Pack	16S1P	15S1P	15S1P
Number of battery	15PCS（15S）	45PCS（15S3P）	75PCS（15S5P）
AC Side			
Rated power	100KW	300KW	500KW
Rated grid voltage	400V		
Rated AC current	140A		
Allowable voltage range	360V~440V		
Rated grid frequency	50Hz/60Hz		
Power Factor	＞0.99		
AC access method	3W+N+PE		
System parameters			
System efficiency	>86%		
Cycle life	＞6000(25℃ 0.5C ＞90%DOD 70%EOL)		
Cooling method	Air Cooling		
Allowable ambient temp.	-30~50℃		
Maximum working altitude	4000m(＞2000mDerating is required)		
Fire mode	Heptafluoropropane gas fire extinguishing device		
Degree of protection	IP54		
Communication method	RS485/CAN		
Operating mode	Microgrid, UPS, Backup power		



NATIONAL BATTERY SUPPLY

Household energy storage products use high-quality square aluminum shell lithium iron phosphate cells, with intelligent battery management system(BMS), Long cycle life, High safety performance, Beautiful appearance, Free combination, and convenient installation.
The battery pack is equipped with LCD display for visualization of operating data. Compatible with most brands solar inverter and be able to communicate with it. It is widely used in distributed PV systems and home energy storage systems. Provide efficient and clean energy for household, industrial, commercial, agricultural and other electrical equipment.

Technical Parameters

Model	L48100	
Specification	48V100Ah	51.2V100Ah
Combination	15S1P	16S1P
Capacity	4.8KWh	5.12KWh
Standard discharge current	50A	50A
Max. discharge current	100A	100A
Working voltage range	40.5-54VDC	43.2-57.6VDC
Standard Voltage	48VDC	51.2VDC
Max. charging current	50A	50A
Max. charging voltage	54V	57.6V
Cycle	3000~6000cycles @DOD 80%/25℃/0.5C	
Working humidity	65±20%RH	
Operating temperature	-10~+50℃	
Working altitude	≤2500m	
Cooling method	Natural cooling	
Installation	Stack installation	
Protection level	IP20	
Maxof parallel	15PCS	
Warranty	5~10 Years	
Communication	Default：RS485/RS232/CAN Optional：WiFi/4G/Bluetooth	
Certified	CE ROHS FCC UN38.3 MSDS	
Product Size	400*200*460mm	
Package Size	460*230*545mm	
Net weight	50kg	53kg
Gross weight	55kg	58kg

*Voltage, capacity, size/color customization, OEM/ODM services can be provided according to customer needs

LOW VOLTAGE PROTECTION
OVER VOLTAGE PROTECTION
OVER CURRENT PROTECTION
OVER TEMPERATURE PROTECTION
BALANCED PROTECTION
OVER DISCHARGE PROTECTION
SHORT CIRCUIT PROTECTION

Rack/Cabinet
Energy Storage Lithium Battery

4U STANDARD | FREE | MAX 15PCS PARALLEL | COMMUNICATION | OEM/ODM



NATIONAL BATTERY SUPPLY

Cabinet series adopts high-quality lithium iron phosphate batteries, equipped with intelligent BMS battery management system, long cycle life, high safety performance, beautiful appearance, free combination and convenient installation. LCD display, visualization of battery operating data. Compatible with most solar inverters, providing efficient energy for photovoltaic off-grid households, commercial and other electrical equipment.

Model	4850		48100		48200	
Specification	48V50Ah	51. 2V50Ah	48V100Ah	51. 2V100Ah	48V200Ah	51. 2V200Ah
Combination	15S1P	16S1P	15S1P	16S1P	15S1P	16S1P
Capacity	2.4KWh	2.56KWh	4.8KWh	5.12KWh	9.6KWh	10.24KWh
Standard discharge current	50A	50A	50A	50A	50A	50A
Max. discharge current	100A	100A	100A	100A	100A	100A
Working voltage range	40.5-54VDC	43.2-57.6VDC	40. 5-54VDC	43.2-57.6VDC	40. 5-54VDC	43.2-57.6VDC
Standard Voltage	48VDC	51.2VDC	48VDC	51.2VDC	48VDC	51.2VDC
Max. charging current	50A	50A	50A	50A	100A	100A
Max. charging voltage	54V	57.6V	54V	57.6V	54V	57.6V
Cycle	3000~6000cycles @DOD 80%/25℃/0.5C					
Working humidity	65±20%RH					
Operating temperature	-10~+50℃					
Working altitude	≤2500m					
Cooling method	Natural cooling					
Installation	Cabinet/Rack Mount					
Protection level	IP20					
Maxof parallel	15PCS					
Warranty	5~10 Years					
Communication	Default：RS485/RS232/CAN Optional：WiFi/4G/Bluetooth					
Certified	CE ROHS FCC UN38.3 MSDS					
Product Size	440*495*88mm		440*495*178mm		440*590*178mm	
Package Size	555*500*130mm		555*500*235mm		480*650*220mm	
Net weight	25kg	37kg	41kg	45kg	101kg	105kg
Gross weight	30kg	42kg	49kg	55kg	111kg	115kg

*Voltage, capacity, size/color customization, OEM/ODM services can be provided according to customer needs

LOW VOLTAGE PROTECTION | OVER VOLTAGE PROTECTION | OVER CURRENT PROTECTION | OVER TEMPERATURE PROTECTION | BALANCED PROTECTION | OVER DISCHARGE PROTECTION | SHORT CIRCUIT PROTECTION



Wall mount
Energy Storage Lithium Battery

MAX 15PCS

WALL MOUNT | PARALLEL | COMMUNICATION | OEM/ODM

NATIONAL BATTERY SUPPLY

Wall mount series adopts high-quality lithium iron phosphate batteries, equipped with intelligent BMS battery management system, long cycle life, high safety performance, beautiful appearance, free combination and convenient installation. LCD display, visualization of battery operating data. Compatible with most solar inverters, providing efficient energy for photovoltaic off-grid households, commercial and other electrical equipment.

Model	4850		48100		48200	
Specification	48V50Ah	51. 2V50Ah	48V100Ah	51. 2V100Ah	48V200Ah	51. 2V200Ah
Combination	15S1P	16S1P	15S1P	16S1P	15S1P	16S1P
Capacity	2.4KWh	2.56KWh	4.8KWh	5.12KWh	9.6KWh	10.24KWh
Standard discharge current	50A	50A	50A	50A	50A	50A
Max. discharge current	100A	100A	100A	100A	100A	100A
Working voltage range	40.5-54VDC	43.2-57.6VDC	40. 5–54VDC	43.2-57.6VDC	40. 5–54VDC	43.2-57.6VDC
Standard Voltage	48VDC	51.2VDC	48VDC	51.2VDC	48VDC	51.2VDC
Max. charging current	50A	50A	50A	50A	100A	100A
Max. charging voltage	54V	57.6V	54V	57.6V	54V	57.6V
Cycle	3000~6000cycles @DOD 80%/25℃/0.5C					
Working humidity	65±20%RH					
Operating temperature	-10~+50℃					
Working altitude	≤2500m					
Cooling method	Natural cooling					
Installation	Wall mount					
Protection level	IP20					
Max of parallel	15PCS					
Warranty	5~10 Years					
Communication	Default：RS485/RS232/CAN Optional：WiFi/4G/Bluetooth					
Certified	CE ROHS FCC UN38.3 MSDS					
Product Size	400*200*585mm		400*230*585mm		400*230*610mm	
Package Size	500*260*630mm		500*290*630mm		460*250*650mm	
Net weight	35kg	40kg	42kg	46kg	102kg	106kg
Gross weight	40kg	45kg	50kg	54kg	112kg	116kg

*Voltage, capacity, size/color customization, OEM/ODM services can be provided according to customer needs

LOW VOLTAGE PROTECTION | OVER VOLTAGE PROTECTION | OVER CURRENT PROTECTION | OVER TEMPERATURE PROTECTION | BALANCED PROTECTION | OVER DISCHARGE PROTECTION | SHORT CIRCUIT PROTECTION



Wall mount
Energy Storage Lithium Battery

WALL MOUNT | MAX 15PCS PARALLEL | COMMUNICATION | OEM/ODM

NATIONAL BATTERY SUPPLY

Wall mount series adopts high-quality lithium iron phosphate batteries, equipped with intelligent BMS battery management system, long cycle life, high safety performance, beautiful appearance, free combination and convenient installation. LCD display, visualization of battery operating data. Compatible with most solar inverters, providing efficient energy for photovoltaic off-grid households, commercial and other electrical equipment.

Model	4850		48100		48200	
Specification	48V50Ah	51. 2V50Ah	48V100Ah	51. 2V100Ah	48V200Ah	51. 2V200Ah
Combination	15S1P	16S1P	15S1P	16S1P	15S1P	16S1P
Capacity	2.4KWh	2.56KWh	4.8KWh	5.12KWh	9.6KWh	10.24KWh
Standard discharge current	50A	50A	50A	50A	50A	50A
Max. discharge current	100A	100A	100A	100A	100A	100A
Working voltage range	40.5-54VDC	43.2-57.6VDC	40. 5–54VDC	43.2-57.6VDC	40. 5–54VDC	43.2-57.6VDC
Standard Voltage	48VDC	51.2VDC	48VDC	51.2VDC	48VDC	51.2VDC
Max. charging current	50A	50A	50A	50A	100A	100A
Max. charging voltage	54V	57.6V	54V	57.6V	54V	57.6V
Cycle	3000~6000cycles @DOD 80%/25℃/0.5C					
Working humidity	65±20%RH					
Operating temperature	-10~+50℃					
Working altitude	≤2500m					
Cooling method	Natural cooling					
Installation	Wall mount					
Protection level	IP20					
Max of parallel	15PCS					
Warranty	5~10 Years					
Communication	Default：RS485/RS232/CAN Optional：WiFi/4G/Bluetooth					
Certified	CE ROHS FCC UN38.3 MSDS					
Product Size	400*200*585mm		400*230*585mm		400*230*610mm	
Package Size	500*260*630mm		500*290*630mm		460*250*650mm	
Net weight	35kg	40kg	42kg	46kg	102kg	106kg
Gross weight	40kg	45kg	50kg	54kg	112kg	116kg

*Voltage, capacity, size/color customization, OEM/ODM services can be provided according to customer needs

LOW VOLTAGE PROTECTION | OVER VOLTAGE PROTECTION | OVER CURRENT PROTECTION | OVER TEMPERATURE PROTECTION | BALANCED PROTECTION | OVER DISCHARGE PROTECTION | SHORT CIRCUIT PROTECTION

Stacked
Energy Storage Lithium Battery

FREE | MAX 15PCS PARALLEL | COMMUNICATION | OEM/ODM



NATIONAL BATTERY SUPPLY



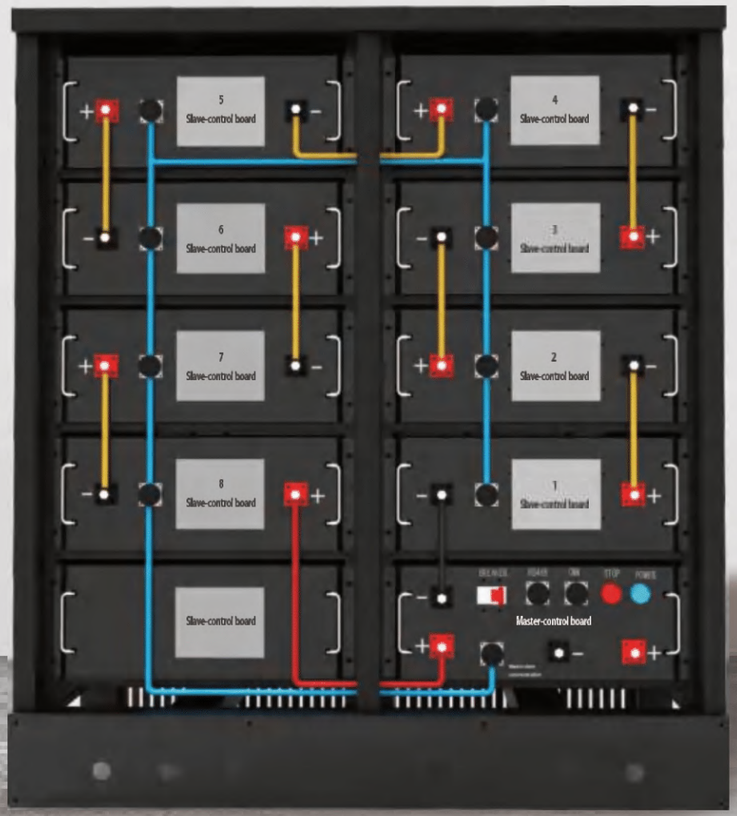
Model	48100		48200	
Specification	48V100Ah	51. 2V100Ah	48V200Ah	51. 2V200Ah
Combination	15S1P	16S1P	15S1P	16S1P
Capacity	4.8KWh	5.12KWh	9.6KWh	10.24KWh
Standard discharge current	50A	50A	50A	50A
Max. discharge current	100A	100A	100A	100A
Working voltage range	40.5-54VDC	43.2-57.6VDC	40. 5-54VDC	43.2-57.6VDC
Standard Voltage	48VDC	51.2VDC	48VDC	51.2VDC
Max. charging current	50A	50A	100A	100A
Max. charging voltage	54V	57.6V	54V	57.6V
Cycle	3000~6000cycles @DOD 80%/25℃/0.5C			
Working humidity	65±20%RH			
Operating temperature	-10~+50℃			
Working altitude	≤2500m			
Cooling method	Natural cooling			
Installation	Stack installation			
Protection level	IP20			
Max of parallel	15PCS			
Warranty	5~10 Years			
Communication	Default：RS485/RS232/CAN Optional：WiFi/4G/Bluetooth			
Certified	CE ROHS FCC UN38.3 MSDS			
Product Size	400*200*460mm		400*240*460mm	
Package Size	460*230*545mm		460*270*545mm	
Net weight	50kg	53kg	102kg	106kg
Gross weight	55kg	58kg	112kg	116kg

*Voltage, capacity, size/color customization, OEM/ODM services can be provided according to customer needs

High voltage
Energy storage lithium battery

96/144/192/240/384/512/768V

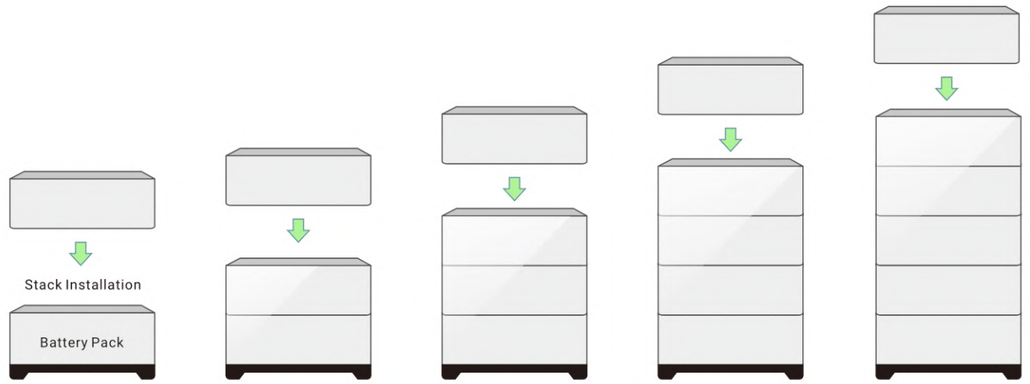
MASTER-SLAVE CONTROL | HIGH VOLTAGE MATCHING | COMMUNICATION | OEM/ODM





NATIONAL BATTERY SUPPLY

High voltage solution supports custom stack form factors



Model	H96	H144	H192	H240	H384	H512	H768
Specification	96V100AH	144V100AH	192V100AH	240V100AH	384V100AH	512V100AH	768V100AH
Capacity	9.6KWh	14.4KWh	19.2KWh	24KWh	38.4KWh	51.2KWh	76.8KWh
Standard discharge current	50A~300A(Customer customize)						
Max discharge current	50A~500A(Customer customize)						
Working voltage range	81-108VDC	121.5-162VDC	162-216VDC	202.5-270VDC	324-432VDC	432-576VDC	648-864VDC
Standard voltage	96V	144V	192V	240V	384V	512V	768V
Max charging current	100A~500A(Customer customize)						
Max charging voltage	81~864V						
Cycle	3000~6000 Cycles @DOD 80%/25℃/0.5C						
Working humidity	65±20%RH						
Operating temperature	-10~+50℃						
Working altitude	≤2500m						
Cooling method	Natural cooling						
Installation	Cabinet/Rack Mount/Stack installation						
Protection level	IP20						
Max of parallel	15PCS						
Warranty	5~10Years						
Communication	Default：RS485/RS232/CAN Optional：WiFi/4G/Bluetooth						
Certified	CE ROHS FCC UN38.3 MSDS						

*Voltage, capacity, size/color customization, OEM/ODM services can be provided according to customer needs

Replace GEL battery
Energy Storage Lithium Battery

FAST | MAX 15PCS PARALLEL | OPTIONAL BLUETOOTH | OPTIONAL GPRS



NATIONAL BATTERY SUPPLY

This series adopts high-quality lithium iron phosphate batteries, equipped with intelligent BMS battery management system, long cycle life, high safety performance, beautiful appearance, free combination and convenient installation. LCD display, visualization of battery operating data. Compatible with most solar inverters, providing efficient energy for photovoltaic off-grid households, commercial and other electrical equipment.

Specification	12V5AH	12V8AH	12V10AH	12V20AH	12V30AH	12V50AH
Capacity	75Wh	102Wh	128Wh	256Wh	384Wh	640Wh
Standard discharge current	2.5A	4A	5A	10A	15A	25A
Max. discharge current	5A	8A	10A	20A	30A	50A
Working voltage range	10-14.6V	10-14.6V	10-14.6V	10-14.6V	10-14.6V	10-14.6V
Standard Voltage	12.8V	12.8V	12.8V	12.8V	12.8V	12.8V
Max. charging voltage	14.6V	14.6V	14.6V	14.6V	14.6V	14.6V
Cycle	3000~6000 cycles@DOD 80%/25℃/0.5C					
Working humidity	65±20%RH					
Operating temperature	-10~+50℃					
Working altitude	≤2500m					
Cooling method	Natural cooling					
Installation	IP65					
Product Size	152*65*98mm	152*65*98mm	152*65*98mm	265*70*90mm	265*70*90mm	255*165*215mm
Net weight	0.9Kg	1.2Kg	1.5Kg	3.2Kg	4.5Kg	9Kg

Specification	12V100AH	12V200AH	12V300AH	24V50AH	24V100AH	24V200AH
Capacity	1.28KWh	2.56KWh	3.84KWh	1.28KWh	2.56KWh	5.12KWh
Standard discharge current	50A	50A	50A	25A	50A	50A
Max. discharge current	100A	100A	100A	50A	100A	100A
Working voltage range	10-14.6V	10-14.6V	10-14.6V	20-29.2V	20-29.2V	20-29.2V
Standard Voltage	12.8V	12.8V	12.8V	25.6V	25.6V	25.6V
Max. charging voltage	14.6V	14.6V	14.6V	29.2V	29.2V	29.2V
Cycle	3000~6000 cycles@DOD 80%/25℃/0.5C					
Working humidity	65±20%RH					
Operating temperature	-10~+50℃					
Working altitude	≤2500m					
Cooling method	Natural cooling					
Installation	IP65					
Product Size	255*165*215mm	480*165*240mm	515*235*225mm	305*165*215mm	480*165*240mm	635*245*220mm
Net weight	16Kg	28Kg	45Kg	20Kg	30Kg	40Kg

L-ESS

Vertical
Lithium Battery Energy Storage System

Li-FePO4 | 3-8KW Out put | APP Control | Customized High/low voltage | Customized capacity



NATIONAL BATTERY SUPPLY

The ESS series product (Energy Storage System) is a high-performance off-grid energy storage system independently developed by us. It uses high-quality lithium iron phosphate batteries and is equipped with an intelligent BMS battery management system. It has long cycle life, high safety performance, and good sealing; High-frequency off-grid photovoltaic inverter, built-in MPPT controller, can provide efficient and reliable energy solutions for photovoltaic off-grid power generation systems, energy storage systems, household photovoltaic storage systems, and industrial and commercial energy storage systems.

The system is equipped with a self-developed APP, supports IOS/Android mobile phones, remotely controls the charging and discharging of the battery pack, monitors the system operating data in real time, and quickly enters the troubleshooting work when the system fails, so as to restore the effective power supply more efficiently.

Model	L-ESS-10	L-ESS-15	L-ESS-20
Capacity	10.24KWh/5KW	15.36KWh/5KW	20.48KWh/5KW
Standard discharge current	50A	50A	50A
Max. discharge current	100A	100A	100A
Working voltage range	43.2-57.6VDC	43.2-57.6VDC	43.2-57.6VDC
Standard Voltage	51.2VDC	51.2VDC	51.2VDC
Max. charging current	50A	50A	50A
Max. charging voltage	57.6V	57.6V	57.6V
Rated PV input voltage	360VDC		
MPPT tracking voltage range	120V-450V		
Max input voltage (VOC) at lowest temperature	500V		
Max input power	6000W		
Number of MPPT tracking paths	1Path		
DC input voltage range	42-60VDC		
Rated mains power input voltage	220VAC/230VAC/240VAC		
Grid power input voltage range	170VAC~280VAC (UPS mode) / 120VAC~280VAC (Inverter mode)		
Grid input frequency range	45Hz~55Hz (50Hz); 55Hz~65Hz (60Hz)		
Inverter output efficiency	94%(MAX)		
Inverter output voltage	220VAC±2%/230VAC±2%/240VAC±2%(Inverter mode)		
Inverter output frequency	50Hz±0.5 or 60Hz±0.5(Inverter mode)		
Inverter output waveform	Pure sine wave		
Grid output efficiency	>99%		
Max mains charging current	60A		
Max PV charging current	100A		
Max charging current (Grid+PV)	100A		
Optional mode	Grid priority/PV priority/Battery priority		
Warranty	5~10Years		
Communication	Optional：RS485/RS232/CAN WiFi/4G/Bluetooth		

*Voltage, capacity, size/color customization, OEM/ODM services can be provided according to customer needs

ESS

Rack/Cabinet
Lithium Battery Energy Storage System

Li-FePO4 | 3-8KW Out put | APP Control | Customized High/low voltage | Customized capacity



NATIONAL BATTERY SUPPLY

The ESS system is a high-performance off-grid energy storage system independently developed by Dingke Energy. It uses high-quality lithium iron phosphate cells and is equipped with an intelligent BMS battery management system. It has long cycle life, high safety performance, good sealing, and is equipped with high-frequency off-grid the inverter, built-in MPPT controller, provides efficient and reliable energy solutions for PV off-grid power generation systems, homes PV storage systems, and industrial and commercial energy storage systems.

The system is equipped with a self-developed APP, supports IOS/Android mobile phones, remotely controls the charging and discharging of the ESS system, and real-time system operation data. When the system fails, it can be quickly eliminated and the effective power supply can be restored efficiently.

Model	C-ESS-5	C-ESS-10	C-ESS-15	C-ESS-20
Capacity	5.12KWh/5KW	10.24KWh/5KW	15.36KWh/5KW	20.48KWh/5KW
Standard discharge current	50A	50A	50A	50A
Max. discharge current	100A	100A	100A	100A
Working voltage range	43.2-57.6VDC	43.2-57.6VDC	43.2-57.6VDC	43.2-57.6VDC
Standard Voltage	51.2VDC	51.2VDC	51.2VDC	51.2VDC
Max. charging current	50A	50A	50A	50A
Max. charging voltage	57.6V	57.6V	57.6V	57.6V
Rated PV input voltage	360VDC			
MPPT tracking voltage range	120V-450V			
Max input voltage (VOC) at lowest temperature	500V			
Max input power	6000W			
Number of MPPT tracking paths	1Path			
DC input voltage range	42-60VDC			
Rated mains power input voltage	220VAC/230VAC/240VAC			
Grid power input voltage range	170VAC~280VAC (UPS mode) / 120VAC~280VAC (Inverter mode)			
Grid input frequency range	45Hz~55Hz (50Hz); 55Hz~65Hz (60Hz)			
Inverter output efficiency	94%(MAX)			
Inverter output voltage	220VAC±2%/230VAC±2%/240VAC±2%(Inverter mode)			
Inverter output frequency	50Hz±0.5 or 60Hz±0.5(Inverter mode)			
Inverter output waveform	Pure sine wave			
Grid output efficiency	>99%			
Max mains charging current	60A			
Max PV charging current	100A			
Max charging current (Grid+PV)	100A			
Optional mode	Grid priority/PV priority/Battery priority			
Warranty	5~10Years			
Communication	Optional: RS485/RS232/CAN WiFi/4G/Bluetooth			

*Voltage, capacity, size/color customization, OEM/ODM services can be provided according to customer needs

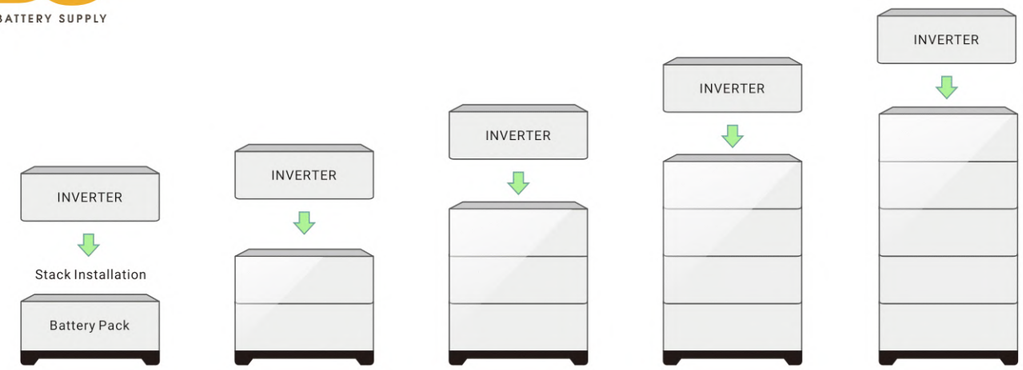
ESS

Stack
Lithium Battery Energy Storage System

Li-FePO4 | 3-8KW Out put | APP Control | Customized High/low voltage | Customized capacity



NATIONAL BATTERY SUPPLY



Model	S-ESS-5	S-ESS-10	S-ESS-15	S-ESS-20
Capacity	5.12KWh/5KW	10.24KWh/5KW	15.36KWh/5KW	20.48KWh/5KW
Standard discharge current	50A	50A	50A	50A
Max. discharge current	100A	100A	100A	100A
Working voltage range	43.2-57.6VDC	43.2-57.6VDC	43.2-57.6VDC	43.2-57.6VDC
Standard Voltage	51.2VDC	51.2VDC	51.2VDC	51.2VDC
Max. charging current	50A	50A	50A	50A
Max. charging voltage	57.6V	57.6V	57.6V	57.6V
Rated PV input voltage	360VDC			
MPPT tracking voltage range	120V-450V			
Max input voltage (VOC) at lowest temperature	500V			
Max input power	6000W			
Number of MPPT tracking paths	1Path			
DC input voltage range	42-60VDC			
Rated mains power input voltage	220VAC/230VAC/240VAC			
Grid power input voltage range	170VAC~280VAC (UPS mode) / 120VAC~280VAC (Inverter mode)			
Grid input frequency range	45Hz~55Hz (50Hz); 55Hz~65Hz (60Hz)			
Inverter output efficiency	94%(MAX)			
Inverter output voltage	220VAC±2%/230VAC±2%/240VAC±2%(Inverter mode)			
Inverter output frequency	50Hz±0.5 or 60Hz±0.5(Inverter mode)			
Inverter output waveform	Pure sine wave			
Grid output efficiency	>99%			
Max mains charging current	60A			
Max PV charging current	100A			
Max charging current (Grid+PV)	100A			
Optional mode	Grid priority/PV priority/Battery priority			
Warranty	5~10Years			
Communication	Optional：RS485/RS232/CAN WiFi/4G/Bluetooth			

*Voltage, capacity, size/color customization, OEM/ODM services can be provided according to customer needs

On/off-grid Solar Inverter

On/off-grid Pure Sine Wave MPPT Communication



Inverter has the function of segmented charging and discharging,backflow prevention and grid connection function.working without battery under special conditions.lithium battery activation function, which can be triggered by mains or PV.Four charging modes: PV Only, Mains Priority, PV Priority and PV&Mains hybrid charging.Two output modes: mains bypass and inverter output, with UPS function.Adopting full-digital double closed loop control, combined with advanced SPWM.Advanced MPPT technology with efficiency of 99.9%.LCD screen and three LED lights can clearly indicate the status and data.Intelligent and variable-speed fan, efficiently dissipate heat and extend system life.Multiple safety protection functions: short circuit protection, overload protection, reverse polarity protection and so on.

Technical Parameters

PRODUCT BASIC PARAMETERS	
Model	PI Series
Rated power	5500W
Standard battery voltage	Lead-acid Battery or Lithium Battery 48VDC
Battery voltage range	100A
Hybrid charging MAX charging current	40VDC~60VDC ± 0.6VDC(Undervoltage/Shutdown Voltage/Overvoltage /Overvoltage Recovery Warning)
Installation method	Wall mount
PV INPUT PARAMETERS	
MaxPV Open-circuit Voltage	500VDC
PV Working Voltage Range	120-500VDC
MPPT Voltage Range	120-450VDC
Max PV Input Current	22A
Max PV Input Power	600W
Max PV Charging Current	100A
AC INPUT PARAMETERS	
Mains Max Charging Current	60A
Rated Input Voltage	220/260VAC
Input Voltage Range	UPS Mains Mode：(170VAC~280VAC)±2% APL Generator Mode：(90VAC-280VAC)±2%
Frequency	50Hz/ 60Hz (Automatic Detection)
Mains Charging Efficiency	＞95%
Switch Time (bypass and inverter)	10ms(Typical Value)
Max Bypass Overload Current	40A
AC OUTPUT PARAMETERS	
Output Waveform	Pure Sine Wave
Rated Output Voltage(VAC)	230VAC 200/208/220/240VAC
Rated Output Power(VA)	6600VA
Rated Output Powe(W)	5500W
Peak Power	11000W(1~3s)
On-load Motor Capacity	4HP
Output Frequency Range(Hz)	50Hz±0.3Hz/60Hz±0.3Hz
Maximum Efficiency	>90%
No-load Loss	Non Energy-saving Mode: ≤50W Energy-saving Mode：≤25W (Manual Setup)
BASIC PARAMETERS	
Working Temperature Range	-25°C ~ 55°C
Storage Temperature Range	-25°C ~ 60°C
Humidity Range	0~100%
Waterproof Grade	IP65
Product Size	556*345*182mm
Product Weight	19.2Kg

Lithium battery
Communication

Rack/Cabinet Solar Inverter

Pure Sine Wave | MPPT | RS485/232 CAN | Standard



NATIONAL BATTERY SUPPLY

PARAMETER			
Model		PC3200	PC5000
Rated power		3200W	5000W
Standard voltage		24VDC	48VDC
Installation		Cabinet/rack installation	
PV PARAMETER			
Working model		MPPT	
Rated PV input voltage		360VDC	
MPPT tracking voltage range		120-450V	
Max input voltage (VOC) at lowest temperature		500V	
Max input power		4000W	6000W
Number of MPPT tracking paths		1 Path	
INPUT			
DC input voltage range		21-30VDC	42-60VDC
Rated mains power input voltage		220/230/240VAC	
Grid power input voltage range		170~280VAC(UPS model)/120~280VAC(inverter model)	
Grid input frequency range		40~55Hz(50Hz) 55~65Hz(60Hz)	
OUTPUT			
Inverter	Output efficiency	94%	
	Output voltage	220VAC±2%/230VAC±2%/240VAC±2%(Inverter model)	
	Output frequency	50Hz±0.5 or 60Hz±0.5(Inverter model)	
Grid	Output efficiency	⩾99%	
	Output voltage range	Following input	
	Output frequency range	Following input	
Battery mode no-load loss		⩽1%(At rated power)	
Grid mode no-load loss		⩽0.5% Rated power(the charger of grid power doesn't work)	
BATTERY			
Battery type	Lead acid battey	Equalizing charging 13.8V Floating charging 13.7V (single battery voltage)	
	Customized battery	The parameter can be set according to customers' requirement (Use different types of battery by setting the panel)	
Max mains charging current		60A	
Max PV charging current		100A	
Max charging current(Grid+PV)		100A	
Charging method		Three-stage (constant current, constant voltage, float charge)	
PROTECTED MODE			
Battery low voltage range		Battery low voltage protection value +0.5V (Single battery voltage)	
Battery voltage protection		Factory default: 10.5V (Single battery voltage)	
Battery over voltage alarm		Equal charging voltage +0.8V (Single battery voltage)	
Battery over voltage protection		Factory default: 17V (Single battery voltage)	
Battery over voltage recovery Voltage		Battery over voltage protection value -1V (Single battery voltage)	
Overload /short circuit protection		Automatic protection (battery mode), circuit breaker or fuse (Grid mode)	
Temperature protection		≥90℃ off output	
PERFORMANCE PARAMETERS			
Conversion time		⩽4ms	
Cooling method		Intelligent cooling fan	
Working temperature		-10~40℃	
Storage temperature		-15~60℃	
Altitude		2000m(>2000m altitude need derating)	
Humidity		0~95% (No condensation)	
Product Size		440*495*178mm	440*495*178mm
Package Size		486*370*198mm	526*384*198mm
Net weight		8.5kg	9.5kg
Gross weight		9.5kg	10.5kg



Lithium battery
Communication

Wall-mount Solar Inverter

Pure Sine Wave | MPPT | RS485/232 CAN | Wall-mounted

NATIONAL BATTERY SUPPLY

PARAMETER			
Model		PW3200	PW5000
Rated power		3200W	5000W
Standard voltage		24VDC	48VDC
Installation		Wall mount installation	
PV PARAMETER			
Working model		MPPT	
Rated PV input voltage		360VDC	
MPPT tracking voltage range		120-450V	
Max input voltage (VOC) at lowest temperature		500V	
Max input power		4000W	6000W
Number of MPPT tracking paths		1 Path	
INPUT			
DC input voltage range		21-30VDC	42-60VDC
Rated mains power input voltage		220/230/240VAC	
Grid power input voltage range		170~280VAC(UPS model)/120~280VAC(inverter model)	
Grid input frequency range		40~55Hz(50Hz) 55~65Hz(60Hz)	
OUTPUT			
Inverter	Output efficiency	94%	
	Output voltage	220VAC±2%/230VAC±2%/240VAC±2%(Inverter model)	
	Output frequency	50Hz±0.5 or 60Hz±0.5(Inverter model)	
Grid	Output efficiency	⩾99%	
	Output voltage range	Following input	
	Output frequency range	Following input	
Battery mode no-load loss		⩽1%(At rated power)	
Grid mode no-load loss		⩽0.5% Rated power(the charger of grid power doesn't work)	
BATTERY			
Battery type	Lead acid battey	Equalizing charging 13.8V Floating charging 13.7V (single battery voltage)	
	Customized battery	The parameter can be set according to customers' requirement (Use different types of battery by setting the panel)	
Max mains charging current		60A	
Max PV charging current		100A	
Max charging current(Grid+PV)		100A	
Charging method		Three-stage (constant current, constant voltage, float charge)	
PROTECTED MODE			
Battery low voltage range		Battery low voltage protection value +0.5V (Single battery voltage)	
Battery voltage protection		Factory default: 10.5V (Single battery voltage)	
Battery over voltage alarm		Equal charging voltage +0.8V (Single battery voltage)	
Battery over voltage protection		Factory default: 17V (Single battery voltage)	
Battery over voltage recovery Voltage		Battery over voltage protection value -1V (Single battery voltage)	
Overload /short circuit protection		Automatic protection (battery mode), circuit breaker or fuse (Grid mode)	
Temperature protection		≥90℃ off output	
PERFORMANCE PARAMETERS			
Conversion time		⩽4ms	
Cooling method		Intelligent cooling fan	
Working temperature		-10~40℃	
Storage temperature		-15~60℃	
Altitude		2000m(>2000m altitude need derating)	
Humidity		0~95% (No condensation)	
Product Size		420*290*110mm	460*304*110mm
Package Size		486*370*198mm	526*384*198mm
Net weight		8. 5kg	9. 5kg
Gross weight		9. 5kg	10. 5kg



Solar Inverter
PLUS+

Lithium battery Communication

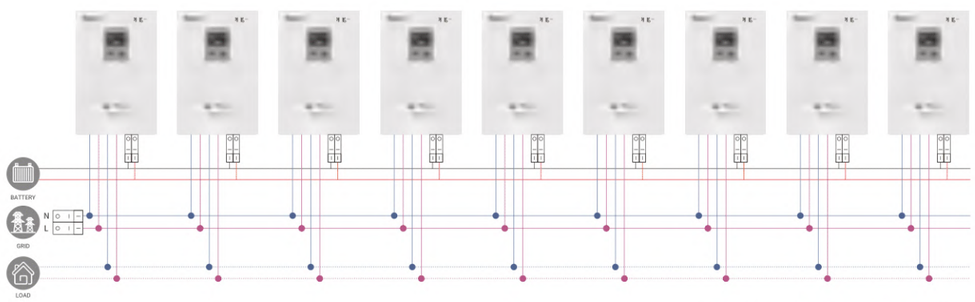
Parallel Use

Pure Sine Wave

MPPT

RS485/232 CAN

Parallel use



Max 9 parallel Max power 45KW

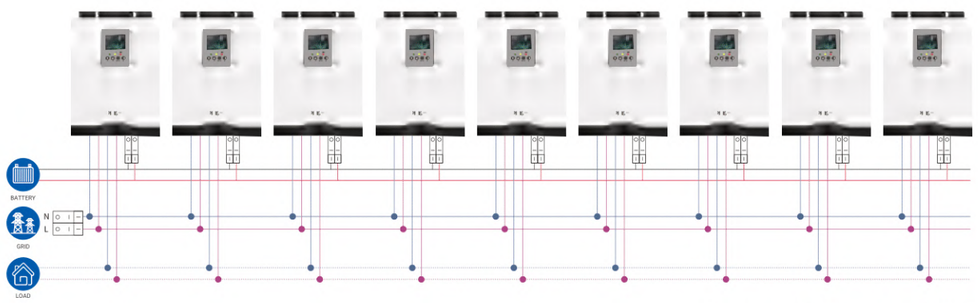
NATIONAL BATTERY SUPPLY

PARAMETER			
Model		PC 5000plus	PW 5000plus
Rated power		5000W	
Standard voltage		48VDC	
Installation		Cabinet/rack installation	Wall mount installation
PV PARAMETER			
Working model		MPPT	
Rated PV input voltage		360VDC	
MPPT tracking voltage range		120-430V	
Max input voltage (VOC) at lowest temperature		450V	
Max input power		5500W	
Number of MPPT tracking paths		1 Path	
INPUT			
DC input voltage range		42-60VDC	
Rated mains power input voltage		208/220/230/240VAC	
Grid power input voltage range		90~280VAC(UPS model)/170~280VAC(inverter model)	
Grid input frequency range		40~70Hz	
OUTPUT			
Inverter	Output efficiency	94%	
	Output voltage	208VAC±2%/220VAC±2%/230VAC±2%/240VAC±2%(Inverter model)	
	Output frequency	50Hz±0.5 or 60Hz±0.5(Inverter model)	
Grid	Output efficiency	⩾99%	
	Output voltage range	Following input	
	Output frequency range	Following input	
Battery mode no-load loss		⩽1%(At rated power)	
Grid mode no-load loss		⩽0.5% Rated power(the charger of grid power doesn't work)	
BATTERY			
Battery type	Lead acid battey	Equal charging voltage 56.6V Float voltage 54V	
	Customized battery	The parameter can be set according to customers' requirement (Use different types of battery by setting the panel)	
Max mains charging current		60A	
Max PV charging current		80A	
Max charging current(Grid+PV)		80A	
Charging method		Three-stage (constant current, constant voltage, float charge)	
PROTECTED MODE			
Battery low voltage alarm		Factory default setting: 44V	
Battery low voltage protection		Factory default setting: 42V	
Battery Over voltage Protection		61VDC	
Overload power protection		Automatic protection (battery mode), circuit breaker or fuse (mains mode)	
Inverter output short circuit protection		Automatic protection (battery mode), circuit breaker or fuse (mains mode)	
Temperature protection		≥90℃ off output	
PERFORMANCE PARAMETERS			
Parallel quantity		9PCS	
Conversion time		⩽4ms	
Cooling method		Intelligent cooling fan	
Working temperature		-10~40℃	
Storage temperature		-15~60℃	
Altitude		2000m(>2000m altitude need derating)	
Humidity		0~95% (No condensation)	
Product Size		420*290*110mm	460*304*110mm
Package Size		486*370*198mm	526*384*198mm
Net weight		8.5kg	9.5kg
Gross weight		9.5kg	10.5kg



Solar Inverter

Pure Sine Wave | MPPT | RS485/232 | Parallel use



Max 9 parallel Max power 45KW

PARAMETER		
Model		PP plus
Rated power		5000W
Standard voltage		48VDC
Installation		Wall mount installation
PV PARAMETER		
Working model		MPPT
Rated PV input voltage		360VDC
MPPT tracking voltage range		120-430V
Max input voltage (VOC) at lowest temperature		450V
Max input power		5500W
Number of MPPT tracking paths		1 Path
INPUT		
DC input voltage range		42-60VDC
Rated mains power input voltage		208/220/230/240VAC
Grid power input voltage range		90~280VAC(UPS model)/170~280VAC(inverter model)
Grid input frequency range		40~70Hz
OUTPUT		
Inverter	Output efficiency	94%
	Output voltage	208VAC±2%/220VAC±2%/230VAC±2%/240VAC±2%(Inverter model)
	Output frequency	50Hz±0.5 or 60Hz±0.5(Inverter model)
Grid	Output efficiency	⩾99%
	Output voltage range	Following input
	Output frequency range	Following input
Battery mode no-load loss		⩽1%(At rated power)
Grid mode no-load loss		⩽0.5% Rated power(The charger of grid power doesn't work)
BATTERY		
Battery type	Lead acid battey	Equal charging voltage 56.6V Float voltage 54V
	Customized battery	The parameter can be set according to customers' requirement (Use different types of battery by setting the panel)
Max mains charging current		60A
Max PV charging current		80A
Max charging current(Grid+PV)		80A
Charging method		Three-stage (constant current, constant voltage, float charge)
PROTECTED MODE		
Battery low voltage alarm		Factory default setting: 44V
Battery low voltage protection		Factory default setting: 42V
Battery Over voltage Protection		61VDC
Overload power protection		Automatic protection (battery mode), circuit breaker or fuse (mains mode)
Inverter output short circuit protection		Automatic protection (battery mode), circuit breaker or fuse (mains mode)
Temperature protection		≥90℃ off output
PERFORMANCE PARAMETERS		
Parallel quantity		9PCS
Conversion time		⩽4ms
Cooling method		Intelligent cooling fan
Working temperature		-10~40℃
Storage temperature		-15~60℃
Altitude		2000m(>2000m altitude need derating)
Humidity		0~95% (No condensation)
Product Size		440*300*110mm
Package Size		515*375*205mm
Net weight		9.5kg
Gross weight		10.5kg



Solar Inverter

Pure Sine Wave | MPPT | RS485/232

NATIONAL BATTERY SUPPLY

PARAMETER									
Model		1000	1500	2000	3200A	3200B	5000A	5000B	7200
Rated power		1000W	1500W	2000W	3200W		5000W		7200W
Standard voltage		12VDC	24VDC				48VDC		
Installation		Wall mount installation							
光伏参数									
Working model		MPPT							
Rated PV input voltage		15-80VDC	30-100VDC			120-450VDC	60-160VDC	120-500VDC	
MPPT tracking voltage range		15-30VDC	30-60VDC			360VDC	60-90VDC	360VDC	
Max input voltage (VOC) at lowest temperature		120VDC				500VDC	180VDC	500VDC	
Max input power		840W	1680W			4000W	3360W	6000W	9000W
Number of MPPT tracking paths		1 Path							2 Path
INPUT									
DC input voltage range		21-30VDC					42-60VDC		
Rated mains power input voltage		220/230/240VAC							
Grid power input voltage range		170~280VAC(UPS Mode)/120~280VAC(Inverter Mode)							
Grid input frequency range		45~55(50Hz) 55~65Hz(60Hz)							
OUTPUT									
Inverter	Output efficiency	94%							
	Output voltage	220VAC±2%/230VAC±2%/240VAC±2%							
	Output frequency	50Hz±0.5 or 60Hz±0.5							
Grid	Output efficiency	>99%							
	Output voltage range	Following input							
	Output frequency range	Following input							
Battery mode no-load loss		⩽1%(At rated power)							
Grid mode no-load loss		⩽0.5% Rated power(The charger of grid power doesn't work)							
BATTERY									
Battery type	Lead acid battey	Equal charging voltage 56.6V Float voltage 54V							
	Customized battery	The parameter can be set according to customers' requirement(Use different types of battery by setting the panel)							
Max mains charging current		120A	100A	110A	120A	100A	120A	120A	150A
Max PV charging current		60A	60A	60A	60A	100A	60A	100A	150A
Max charging current(Grid+PV)		60A	40A	50A	60A	60A	60A	60A	80A
Charging method		Three-stage (constant current, constant voltage, float charge)							
PROTECTED MODE									
Battery low voltage alarm		Battery low voltage protection value +0.5V (Single battery voltage)							
Battery voltage protection		Factory default: 10.5V (Single battery voltage)							
Battery over voltage alarm		Equal charging voltage +0.8V (Single battery voltage)							
Battery over voltage protection		Factory default: 17V (Single battery voltage)							
Battery over voltage recovery		Battery over voltage protection value -1V (Single battery voltage)							
Overload/short circuit protection		Automatic protection (Battery mode), circuit breaker or fuse (Grid mode)							
Temperature protection		≥90℃ off output							
PERFORMANCE PARAMETERS									
Conversion time		⩽10ms							
Cooling method		Intelligent cooling fan							
Working temperature		-10~40℃							
Storage temperature		-15~60℃							
Altitude		2000m(>2000m altitude need derating)							
Humidity		0~95% (No condensation)							
Product Size		355*272*91.5mm			400*315*101mm		440*342*101mm		525*352*115mm
Package Size		443*350*187mm			488*393*198mm		528*420*198mm		615*435*210mm
Net weight		6.5kg			8.2kg		10kg		14kg
Gross weight		7.5kg			9.5kg		11kg		15.5kg



Three-phase Low Frequency Inverter

Pure Sine Wave | MPPT | Three-phase | RS485/232

NATIONAL BATTERY SUPPLY

Model:		32248/96/192	48248/96/192	56248/96/192	64248/96/192	80248/96/192	10348/96/192	12348/96/192	16396/192	20396/192	24396/192
Rated Power		4KVA/3.2KW	6KVA/4.8KW	7KVA/5.6KW	8KVA/6.4KW	10KVA/8KW	12.5KVA/10KW	15KVA/12KW	20KVA/16KW	25KVA/20KW	30KVA/24KW
Peak Power(20ms)		9.6KVA	14.4KVA	16.8KVA	19.2KVA	24KVA	30KVA	36KVA	48KVA	60KVA	72KVA
Start Motor		3HP	4HP	4HP	4HP	5HP	6HP	7HP	10HP	10HP	15HP
Battery Voltage		48/96/192VDC							96/192VDC		
Built-in solar controller charging current (Optional)		PWM:10A-60A(48V System);50A/100A(96V System);50A(192V System) MPPT:10A-100A(48V System);50A/100A(96V System)				PWM: 50A/100A (96V System); 50A/100A (192V System) MPPT: 50A/100A (96V System)					
Size(L*W*Hmm)		565*300*775				725*365*1010					
Package Size(L*W*Hmm)		625*360*895				785*425*1135					
N.W.(kg)		65	73	75	80	112	122	134	160	176	189
G.W.(kg)(Wooden Packing)		78	86	88	93	136	146	158	184	200	213
Installation Method		Tower									

Model:		323192	403192	483384	643384	803384	963384	1003384	1203384	1283384	1503384	1603384
Rated Power		40KVA/32KW	50KVA/40KW	60KVA/48KW	80KVA/64KW	100KVA/80KW	120KVA/96KW	125KVA/100KW	150KVA/120KW	160KVA/128KW	190KVA/150KW	200KVA/160KW
Peak Power(20ms)		96KVA	120KVA	144KVA	192KVA	240KVA	288KVA	300KVA	360KVA	384KVA	450KVA	480KVA
Start Motor		15HP	20HP	25HP	30HP	40HP	50HP	50HP	60HP	60HP	80HP	80HP
Battery Voltage		192VDC		384VDC								
Built-in solar controller charging current (Optional)		PWM:100A-200A(192V&384V System) MPPT:50A/100A(192V&384V System)		PWM: 100A-200A / MPPT: 50A/100A								
Size(L*W*Hmm)		720*575*1275		875*720*1380							1123*900*1605	
Package Size(L*W*Hmm)		785*640*1400		980*825*1560							1185*960*1750	
N.W.(kg)		240	260	290	308	512	542	552	612	642	705	755
G.W.(kg)(Wooden Packing)		273	293	323	341	552	582	592	652	692	755	805
Installation Method		Tower										
Input	DC Input Voltage Range	10.5-15VDC (Single battery voltage)										
	AC Input Voltage Range	380Vac/400Vac -85%~+120%(customized 190Vac/200Vac)										
	AC Input Frequency Range	45Hz-55Hz (50Hz) / 55Hz-65Hz (60Hz)										
	Max AC charging current	0~45A (Depending on the model)										
	AC charging method	Three-stage (constant current, constant voltage, floating charge)										
	Phase	3/N/PE										
Output	Efficiency(Battery Mode)	≥85%										
	Output Voltage(Battery Mode)	380Vac/400Vac±2%(customized 190Vac/200Vac)										
	Output Frequency(Battery Mode)	50/60Hz±1%										
	Output Wave(Battery Mode)	Pure Sine Wave										
	Output waveform distortion	Linear load≤3%										
	Efficiency(AC Mode)	>99%										
	Output Voltage(AC Mode)	Conforming to AC input										
	Output Frequency(AC Mode)	Conforming to AC input										
	No load loss(Battery Mode)	≤2.5% rated power (4KVA-30KVA models) ; ≤1% rated power (40KVA-200KVA models)										
	No load loss(AC Mode)	≤2% rated power(charger does not work in AC mode)										
	No load loss(Energy saving Mode)	≤10W										
	Phase	3/N/PE										
Battery Type	VRLA Battery	Charge Voltage :13.8V; Float Voltage:13.7V (Single battery voltage)										
	Customize battery	Charging and discharging parameters of different types of batteries can be customized according to user requirements (charging and discharging parameters of different types of batteries can be set through the operation panel)										
Protection	Battery undervoltage alarm	11V (Single battery voltage)										
	Battery undervoltage protection	10.5V (Single battery voltage)										
	Battery overvoltage alarm	15V (Single battery voltage)										
	Battery overvoltage protection	17V (Single battery voltage)										
	Battery overvoltage recovery voltage	14.5V (Single battery voltage)										
	Overload power protection	Automatic protection (battery mode), circuit breaker or insurance (AC mode)										
	Inverter output short circuit protection	Automatic protection (battery mode), circuit breaker or insurance (AC mode)										
	Temperature protection	>90°C (Shut down output)										
Alarm	A	Normal working condition, buzzer has no alarm sound										
	B	Buzzer sounds 4 times per second when battery failure, voltage abnormality, overload protection										
	C	When the machine is turned on for the first time, the buzzer will prompt 5 when the machine is normal										
Inside Solar controller (Optional)	Charging Mode	MPPT or PWM										
	Charging current	PWM: 10A/20A/30A/40A/50A/60A(48V System); 50A/100A/150A/200A(96V/192V/384V System) MPPT: 10A/20A/30A/40A/50A/60A/80A/100A(48V System); 50A/100A(96V/192V/384V System)										
	PV Input Voltage Range	PWM: 60V-88V(48V System); 120V-176V(96V System); 240V-352V(192V System); 480V-704V(384V System) MPPT: 60V-120V(48V System); 120V-240V(96V System); 240V-360V(192V System); 480V-640V(384V System)										
	Max PV Input Voltage(Voc) (At the lowest temperature)	PWM: 100V(48V System); 200V(96V System); 400V(192V System); 750V(384V System) MPPT: 150V(48V System); 300V(96V System); 450V(192V System); 800V(384V System)										
	PV Array Maximum Power	48V System: 560W(10A)/1120W(20A)/1680W(30A)/2240W(40A)/2800W(50A)/3360W(60A) 96V System: (PWM: 5.6KW(50A)/11.2KW(100A)) / (MPPT: 5.6KW(50A)/5.6KW*2(100A)); 192V System: (PWM: 11.2KW(50A)/22.4KW(100A)/16.8KW*2(150A)/22.4KW*2(200A)) / (MPPT: 11.2KW(50A)/11.2KW*2(100A)); 384V System: (PWM: 22.4KW(50A)/44.8KW(100A)/33.6KW*2(150A)/44.8KW*2(200A)) / (MPPT: 22.4KW(50A)/22.4KW*2(100A))										
	Standby loss	≤3W										
	Maximum conversion efficiency	>95%										
Working Mode		Battery First/AC First/Saving Energy Mode										
Transfer Time		≤4ms										
Display		LCD										
Thermal method		Forced air cooling										
Communication(Optional)		RS485/APP (WIFI monitoring or GPRS monitoring)										
Environment	Operating temperature	-10℃~40℃										
	Storage temperature	-15℃~60℃										
	Noise	≤65dB										
	Elevation	2000m (More than derating)										
	Humidity	0%~95% (No condensation)										
Warranty		1 year										

Note: 1. Specifications are subject to change without prior notice; 2. Special voltage and power requirements can be customized according to the actual situation of users.



Single-phase Low Frequency Inverter

Pure Sine Wave | MPPT | Single-phase | RS485/232

NATIONAL BATTERY SUPPLY

Model:		70112/24 (701)	10212/24 (102)	15224/48 (152)	20224/48 (202)	30224/48 (302)	35248/96 (352)	40248/96 (402)	50248/96 (502)	60248/96 (602)	70248/96/192 (702)
Rated Power		700W	1000W	1500W	2000W	3000W	3500W	4000W	5000W	6000W	7000W
Peak Power (20ms)		2100VA	3000VA	4500VA	6000VA	9000VA	10500VA	12000VA	15000VA	18000VA	21000VA
Start Motor		0.5HP	1HP	1.5HP	2HP	3HP	3HP	3HP	4HP	4HP	5HP
Battery Voltage		12/24VDC	12/24VDC	24/48VDC	24/48VDC	24/48VDC	48/96VDC	48/96VDC	48/96VDC	48/96VDC	48/96/192VDC
Max AC charging current		0A~20A (Depending on the model, The maximum charging power is 1/4 of the rated power)									
Built-in solar controller charging current (optional)		10A~60A(PWM or MPPT)			24/48V(PWM:10A~60A/MPPT:10A-100A)			48V(PWM:10A~120A/MPPT:10A~100A) / 96V(50A/100A(PWM or MPPT))			
Size(L*W*Hmm)		340x165x283			410x200x350			491x260x490			
Packing Size(L*W*Hmm)		405x230x340(1pc) / 475x415x350(2pc)			475x265x410			545x315x550			
N.W.(kg)		9.5(1pc)	10.5(1pc)	11.5(1pc)	17	20.5	21.5	29	30	31.5	36
G.W.(kg) (Carton packaging)		11(1pc)	12(1pc)	13(1pc)	19	22.5	23.5	32	33	34.5	39
Installation Method		Tower									

Model:		80248/96/192 (802)	10348/96/192 (103)	12396/192 (123)	153192 (153)	203192 (203)	253240 (253)	303240 (303)	403384 (403)
Rated Power		8KW	10KW	12KW	15KW	20KW	25KW	30KW	40KW
Peak Power (20ms)		24KVA	30KVA	36KVA	45KVA	60KVA	75KVA	90KVA	120KVA
Start Motor		5HP	7HP	7HP	10HP	12HP	15HP	15HP	20HP
Battery Voltage		48/96/192VDC	48/96V/192VDC	96/192VDC	192VDC	192VDC	240VDC	240VDC	384VDC
Max AC charging current		0A~40A (Depending on the model, The maximum charging power is 1/4 of the rated power)			0A~20A (Depending on the model, The maximum charging power is 1/4 of the rated power)				
Built-in solar controller charging current (optional)		PWM:(48V: 120A; 96V:50A/100A; 192V/384V:50A) MPPT:(48V:100A/200A; 96V:50A/100A; 192V/384V:50A)			50A/100A			50A/100A	
Size(L*W*Hmm)		540x350x695			593x370x820			721x400x1002	
Packing Size(L*W*Hmm)		600*410*810			656*420*937			775x465x1120	
N.W.(kg)		66	70	77	110	116	123	167	192
G.W.(kg) (Wooden packing)		77	81	88	124	130	137	190	215
Installation Method		Tower							
Input	DC Input Voltage Range	10.5-15VDC (Single battery voltage)							
	AC Input Voltage Range	73VAC~138VAC(110VAC)/83VAC~148VAC(120VAC)/145VAC~275VAC(220VAC)/155VAC~285VAC(230VAC)/165VAC~295VAC(240VAC)(700W~7000W) 92VAC~128VAC(110VAC)/102VAC~138VAC(120VAC)/185VAC~255VAC(220VAC)/195VAC~265VAC(230VAC)/205VAC~275VAC(240VAC)(8KW~40KW)							
	AC Input Frequency Range	45Hz~55Hz (50Hz) / 55Hz~65Hz (60Hz)							
	AC charging method	Three-stage (constant current, constant voltage, floating charge)							
Output	Efficiency(Battery Mode)	≥85%							
	Output Voltage(Battery Mode)	110VAC±2% / 120VAC±2% / 220VAC±2% / 230VAC±2% / 240VAC±2%							
	Output Frequency(Battery Mode)	50Hz±0.5 or 60Hz±0.5							
	Output Wave(Battery Mode)	Pure Sine Wave							
	Efficiency(AC Mode)	≥99%							
	Output Voltage(AC Mode)	110VAC±10%/120VAC±10%/220VAC±10%/230VAC±10%/240VAC±10%(For models below or equal to 7KW); Follow Input (For models above 7KW)							
	Output Frequency(AC Mode)	Follow input							
	Output waveform distortion(Battery Mode)	≤3% (Linear load)							
	No load loss(Battery Mode)	≤1% rated power							
	No load loss(AC Mode)	≤2% rated power (charger does not work in AC mode))							
	No load loss(Energy saving Mode)	≤10W							
Battery Type (selectable)	VRLA Battery	Charge Voltage :14.2V; Float Voltage:13.8V(Single battery voltage)							
	Customize battery	Charging and discharging parameters of different types of batteries can be customized according to user requirements (charging and discharging parameters of different types of batteries can be set through the operation panel)							
Protection	Battery undervoltage alarm	Factory default: 11V (Single battery voltage)							
	Battery undervoltage protection	Factory default: 10.5V (Single battery voltage)							
	Battery overvoltage alarm	Factory default: 15V (Single battery voltage)							
	Battery overvoltage protection	Factory default: 17V (Single battery voltage)							
	Battery overvoltage recovery voltage	Factory default: 14.5V (Single battery voltage)							
	Overload power protection	Automatic protection (battery mode), circuit breaker or insurance (AC mode)							
	Inverter output short circuit protection	Automatic protection (battery mode), circuit breaker or insurance (AC mode)							
	Temperature protection	>90°C (Shut down output)							
Alarm	A	Normal working condition, buzzer has no alarm sound							
	B	Buzzer sounds 4 times per second when battery failure, voltage abnormality, overload protection							
	C	When the machine is turned on for the first time, the buzzer will prompt 5 when the machine is normal							
Inside Solar controller (Optional)	Charging Mode	PWM or MPPT							
	PV Input Voltage Range	PWM: 15V-44V(12V system); 30V-44V(24V system); 60V-88V(48V system);120V-176V(96V system); 240V-352V(192V system) ; 300V-400V(240V system); 480V-704V(384V system) MPPT: 15V-120V(12V system); 30V-120V(24V system); 60V-120V(48V system);120V-240V(96V system); 240V-360V(192V system); 300V-400V(240V system);480V-640V(384V system)							
	Max PV Input Voltage(Voc) (At the lowest temperature)	PWM: 50V(12V/24V system); 100V(48V system); 200V(96V system); 400V(192V system); 500V(240V system) ;750V(384V system) MPPT: 150V(12V/24V/48V system); 300V(96V system); 450V(192V system); 500V(240V system) ;800V(384V system)							
	PV Array Maximum Power	12V system: 140W(10A)/280W(20A)/420W(30A)/560W(40A)/700W(50A)/840W(60A)/1120W(80A)/1400W(100A); 24V system: 280W(10A)/560W(20A)/840W(30A)/1120W(40A)/1400W(50A)/1680W(60A)/2240W(80A)/2800W(100A); 48V system: 560W(10A)/1120W(20A)/1680W(30A)/2240W(40A)/2800W(50A)/3360W(60A)/4480W(80A)/5600W(100A)/6720W(PWM 120A)/5.6KW&11.2KW(MPPT 100A/200A); 96V system: 5.6KW(50A)/11.2KW(100A); 192V system: (PWM:11.2KW(50A)/22.4KW(100A)) / (MPPT:11.2KW(50A)/11.2*2KW(100A)); 240V system: (PWM:14KW(50A)/28KW(100A)) / (MPPT:14KW(50A)/14*2KW(100A)); 384V system: (PWM:22.4KW(50A)/44.8KW(100A)) / (MPPT:22.4KW(50A)/22.4*2KW(100A))							
	Standby loss	≤3W							
	Maximum conversion efficiency	>95%							
Working Mode		Battery First/AC First/Saving Energy Mode							
Transfer Time		≤4ms							
Display		LCD							
Thermal method		Cooling fan in intelligent control							
Communication(Optional)		RS485/APP (WIFI monitoring or GPRS monitoring)							
Environment	Operating temperature	-10℃~40℃							
	Storage temperature	-15℃~60℃							
	Noise	≤55dB							
	Elevation	2000m (More than derating)							
	Humidity	0%~95% ,No condensation							
	Warranty	1 year							

Note: 1. Specifications are subject to change without prior notice; 2. Special voltage and power requirements can be customized according to the actual situation of users.

Portable Outdoor Power Supply

600W 576Wh



Portable mobile power supply is a portable energy storage power supply that integrates multiple functions. It has built-in high-performance A-grade lithium iron phosphate batteries, high-quality BMS management system, and high-efficiency inverter conversion circuit. It can be used as a home, office, outdoor camping and emergency backup power supply. Mains or solar power available charging, can provide rated 220V/ 600W AC output, 5V/12V DC output, USB output, Type C, wireless output.

- Small, light weight, convenient
- Pure sine wave output,
- 0.3S quick start, High efficiency
- Support QC3.0 fast charging, support PD18-100W fast charging, 15W wireless charging
- LCD display, visualization of operating data

NATIONAL BATTERY SUPPLY

Model	600				
Rated Capacity	576WH				
Rated power	600W				
Charge and discharge parameters					
Operating temperature	0℃~40℃ (Charge) -10°C~40°C(Discharge)				
Support charging type	Grid charging/Solar charging				
Input voltage range	Grid input 100V/solar input 20V				
Input voltage	Grid input 100V/Solar input 20V				
Charging voltage	12V				
AC Max charging current	50V/5A				
Max charging power	100W				
Output type	AC	USB	Type-C	DC	Wireless charging
Output rated power	600W	QC3.0 18W	100W MAX	100W	15W
Frequency	50Hz/60 Hz±1HZ(Auto)				
Output voltage/current	110~230V/10A	5V/2.4A,5V/3A	5~20V/3.25A	12.5V 8A	12V/1.25A
Product Features					
Cooling method	cooling fan				
Running noise	⩽5dB				
Protection	IP53				
Working humidity	65±20%RH				
Storage humidity	65±20%RH				
Product Information					
Product Size	265*200*195mm				
Package Size	365*325*298mm				
Net weight	4.9kg				
Gross weight	6.5kg				
Packing	Carton				

Protection



Customized sockets



Portable Outdoor Power Supply

1200W 1008Wh



Portable mobile power supply is a portable energy storage power supply that integrates multiple functions. It has built-in high-performance A-grade lithium iron phosphate batteries, high-quality BMS management system, and high-efficiency inverter conversion circuit. It can be used as a home, office, outdoor camping and emergency backup power supply. Mains or solar power available charging, can provide rated 220V/ 1200W AC output, 5V/12V DC output, USB output, Type C, wireless output.

- Small, light weight, convenient
- Pure sine wave output,
- 0.3S quick start, High efficiency
- Support QC3.0 fast charging, support PD18-65W fast charging, 15W wireless charging
- LCD display, visualization of operating data

NBS
NATIONAL BATTERY SUPPLY

Model	1200				
Rated Capacity	1008WH				
Rated power	1200W				
Charge and discharge parameters					
Operating temperature	0℃~40℃ (Charge) -10°C~40°C(Discharge)				
Support charging type	Grid charging/Solar charging				
Input voltage range	Grid input 100V/solar input 20~40V				
Input voltage	Grid input 100V/Solar input 20~40V				
Charging voltage	16.8V				
AC Max charging current	100V/10A				
Max charging power	550W				
Output type	AC	USB	Type-C	DC	Wireless charging
Output rated power	1200W	QC3.0 18W	65W MAX	100W	15W
Frequency	50Hz/60 Hz±1HZ(Auto)				
Output voltage/current	110~230V/10A	5V/2.4A,5V/3A	5~20V/3.25A	12.5V 8A	12V/1.25A
Product Features					
Cooling method	Cooling fan				
Running noise	⩽5dB				
Protection	IP53				
Working humidity	65±20%RH				
Storage humidity	65±20%RH				
Product Information					
Product Size	370*245*275mm				
Package Size	455*325*368mm				
Net weight	13kg				
Gross weight	15kg				
Packing	Carton				

Protection

Low voltage Protection · Over voltage Protection · Over current Protection · Over temperature Protection · Balanced Protection · Over discharge Protection · Short circuit Protection

Customized sockets





2000-3000 PRO

Portable Outdoor Power Supply

Portable mobile power supply is a portable energy storage power supply that integrates multiple functions. It has built-in high-performance A-grade lithium iron phosphate batteries, high-quality BMS management system, and high-efficiency inverter conversion circuit. It can be used as a home, office, outdoor camping and emergency backup power supply. Mains or solar power available charging, can provide rated 220V/3200W AC output, 5V/12V DC output, USB output, Type C, wireless output.

- Large capacity,High power
- Pure sine wave output,
- 0.3S quick start, High efficiency
- Support QC3.0 fast charging, support PD18-65W fast charging, 15W wireless charging
- LCD display, visualization of operating data

NATIONAL BATTERY SUPPLY

Model	2000 PRO	3000 PRO
Rated Capacity	2016Wh	3000WH
Rated power	2000W	3200W
Charge and discharge parameters		
Operating temperature	0℃~40℃ (Charge) -10°C~40°C(Discharge)	
Support charging type	Grid charging/Solar charging	
Input voltage range	Grid input 85-250V/solar input 26~60V	
Input voltage	Grid input 250V/Solar input 26~60V	
Charging voltage	16.8V	
AC Max charging current	100V/10A	
Max charging power	1000W	

Output type	AC	USB	Type-C	DC	Wireless charging
Output rated power	2000~3200W	QC3.0 18W	65W MAX	100W	15W
Frequency	50Hz/60 Hz±1HZ(Auto)				
Output voltage/current	110~230V/10A	5V/2.4A,5V/3A	5~20V/3.25A	12.5V 8A	12V/1.25A

Product Features	
Cooling method	Cooling fan
Running noise	⩽5dB
Protection	IP53
Working humidity	65±20%RH
Storage humidity	65±20%RH
Product Information	
Product Size	400*300*322mm
Package Size	435*355*352mm
Net weight	28kg
Gross weight	32kg
Packing	Carton

Protection

Low voltage Protection · Over voltage Protection · Over current Protection · Over temperature Protection · Balanced Protection · Over discharge Protection · Short circuit Protection

Customized sockets

Multifunction · U.S. · GFCI · Australia · U.K. · Lsrael · Switzerland · France · Germany · Japan · SouthAfrica



Power Supply

3-5KW 3-5KWh

- Large capacity,High power
- Pure sine wave output,
- LCD display, visualization of operating data

NATIONAL BATTERY SUPPLY

PARAMETER			
Model		3000	5000
Rated power		3200W	5000W
Capacity		3KWh	5.12KWh
PV PARAMETER			
Working model		MPPT	
Rated PV input voltage		360VDC	
MPPT tracking voltage range		120-450V	
Max input voltage (VOC) at lowest temperature		500V	
Max input power		4000W	6000W
Number of MPPT tracking paths		1 Path	
INPUT			
DC input voltage range		21-30VDC	42-60VDC
Rated mains power input voltage		220/230/240VAC	
Grid power input voltage range		170~280VAC(UPS model)/120~280VAC(inverter model)	
Grid input frequency range		40~55Hz(50Hz) 55~65Hz(60Hz)	
OUTPUT			
Inverter	Output efficiency	94%	
	Output voltage	220VAC±2%/230VAC±2%/240VAC±2%(Inverter model)	
	Output frequency	50Hz±0.5 or 60Hz±0.5(Inverter model)	
Grid	Output efficiency	⩾99%	
	Output voltage range	Following input	
	Output frequency range	Following input	
Battery mode no-load loss		⩽1%(At rated power)	
Grid mode no-load loss		⩽0.5% Rated power(the charger of grid power doesn't work)	
BATTERY			
Battery type	Lead acid battey	Equalizing charging 13.8V Floating charging 13.7V (single battery voltage)	
	Customized battery	The parameter can be set according to customers' requirement (Use different types of battery by setting the panel)	
Max mains charging current		60A	
Max PV charging current		80A	
Max charging current(Grid+PV)		80A	
Charging method		Three-stage (constant current, constant voltage, float charge)	
PROTECTED MODE			
Battery low voltage alarm		Factory default setting: 44V	
Battery low voltage protection		Factory default setting: 42V	
Battery Over voltage Protection		61VDC	
Overload power protection		Automatic protection (battery mode), circuit breaker or fuse (mains mode)	
Inverter output short circuit protection		Automatic protection (battery mode), circuit breaker or fuse (mains mode)	
Temperature protection		≥90℃ off output	
PERFORMANCE PARAMETERS			
Conversion time		⩽4ms	
Cooling method		Intelligent cooling fan	
Working temperature		-10~40℃	
Storage temperature		-15~60℃	
Altitude		2000m(>2000m altitude need derating)	
Humidity		0~95% (No condensation)	
Product Size		500*300*340mm	500*300*340mm
Package Size		600*400*440mm	600*400*440mm
Net weight		42kg	55kg
Gross weight		48kg	65kg

NATIONAL BATTERY SUPPLY

WE LOOK FORWARD TO HEARING FROM *YOU!*

844-539-2555 (Toll Free)

Sales@NationalBatterySupply.com

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OEM private label high energy density and deep cycle battery manufacture.